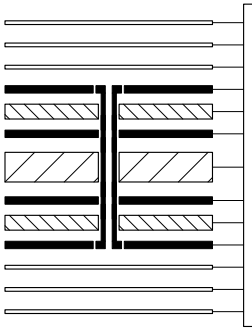
IndustrialSense Fabrication Document

Layer Stack Legend



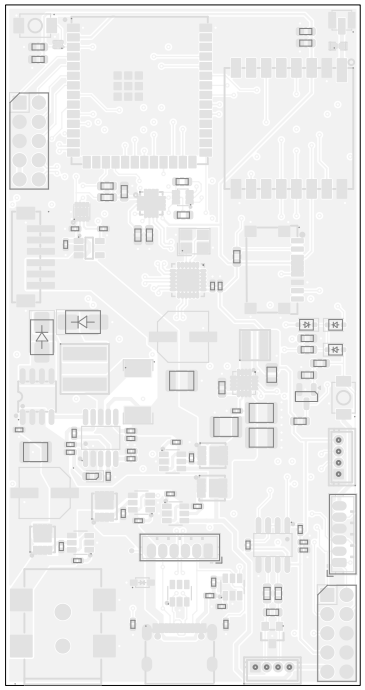
Material	Layer	Thickness	Dielectric	Type	Gerber
	F.Paste			Paste Mask	
	F.Silkscreen		Direct Printing	Legend	GBR
	F.Mask	0.02mm	Solder Resist	Solder Mask	GBR
Copper	**L1**	**0.07mm (2.00oz)**		**Signal**	**GBR**
Prepreg		0.1mm	FR4_7628	Dielectric	
Copper	**L2**	**0.035mm (1.00oz)**		**Signal**	**GBR**
Core		0.75mm	FR4	Dielectric	
Copper	**L3**	**0.035mm (1.00oz)**		**Signal**	**GBR**
Prepreg		0.1mm	FR4_7628	Dielectric	
Copper	**L4**	**0.07mm (2.00oz)**		**Copper Layer**	**GBR**
	B.Mask	0.02mm	Solder Resist	Solder Mask	GBR
	B.Silkscreen		Direct Printing	Legend	GBR
	B.Paste			Paste Mask	

Total thickness: 1.2mm
Note: external layer thicknesses are specified after plating

Impedance Table

Transmission Line	Impedance [ohms]	Tolerance [ohms]	Layer	Trace Width [mm]	Gap [mm]	Ref. Layers
Edge-Coupled Coated Microstrip	100	±10 %	L1	0.2032	0.28	L2

Top Fabrication (Scale 1:1)



FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

1) FABRICATE PER IPC-6012A CLASS 2.

2) OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge_Cuts.GBR" SUFFIX.

   DIMENSIONS OF CIRCUMSIZED RECTANGLE SHOWN ON THIS DRAWING FOR REFERENCE ONLY.

3) SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.

   SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING FOR REFERENCE ONLY.

4) SURFACE FINISH: HAL LEAD-FREE

5) SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR BLACK.

6) SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING YELLOW NON-CONDUCTIVE EPOXY INK.

7) ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDERMASK OPENED IN GERBER.

8) VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING

9) PCB MATERIAL REQUIREMENTS:

   A. FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS.
   B. Tg 170 C OR EQUIVALENT.
   C. EQUIVALENT MATERIAL SHALL BE RoHS COMPLIANT, HALOGEN FREE AND APPROVED BY NARVAL INC..

10) DESIGN GEOMETRY MINIMUM FEATURE SIZES:

BOARD SIZE	46.900 × 90.000 mm
BOARD THICKNESS	1.200 mm
TRACE WIDTH	0.150 mm
TRACE TO TRACE	-0.000 mm
MIN. HOLE (PTH)	0.300 mm
MIN. HOLE (NPTH)	N/A mm
ANNULAR RING	0.140 mm
COPPER TO HOLE	0.175 mm
COPPER TO EDGE	0.500 mm
HOLE TO HOLE	0.254 mm

11) REFER TO IMPEDANCE TABLE FOR IMPEDANCE CONTROL REQUIREMENTS.

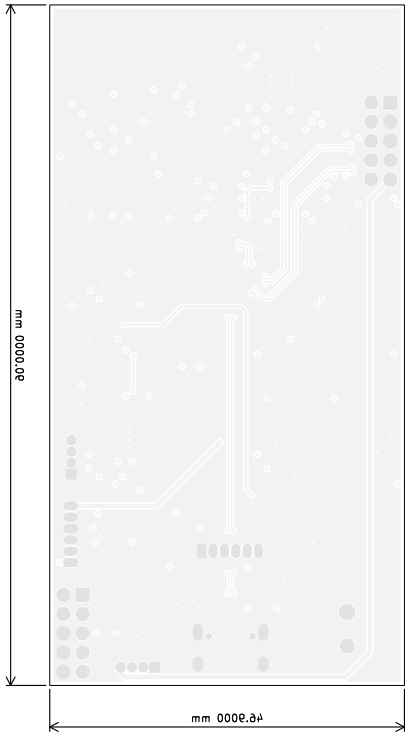
12) CONFIRM SPACE WIDTHS AND SPACINGS.

All dimensions are in millimeters unless otherwise specified.

Comments:	Company: Narval Inc.		Variant: PRELIMINARY	Git Hash: 3a63ce1
	Board Name: IndustrialSense		Project Name: IoT NG	
Sheet Title: Top Fabrication (Scale 1:1)	File Name: IndustrialSense.kicad_pcb	Designer: William Regalado	Date: 2024-04-13	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: A4	Sheet: 1 of 9

IndustrialSense Fabrication Document

Bottom Fabrication (Scale 1:1)



All dimensions are in millimeters unless otherwise specified.

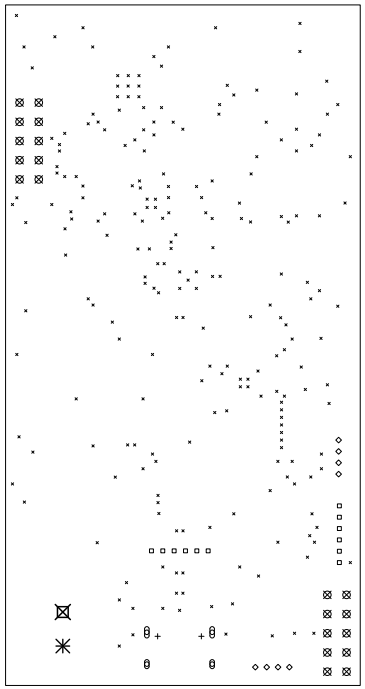
Comments:	Company: Narval Inc.		Variant: PRELIMINARY	Git Hash: 3a63ce1
	Board Name: IndustrialSense		Project Name: IoT NG	
Sheet Title: Bottom Fabrication (Scale 1:1)	File Name: IndustrialSense.kicad_pcb	Designer: William Regalado	Date: 2024-04-13	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: A4	Sheet: 2 of 9

IndustrialSense Fabrication Document

Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	226	0.30mm (11.81mils)	PTH	Round	L1 - L4	Via
○	4	0.60mm (23.62mils)	PTH	Slot	L1 - L4	Pad
+	2	0.70mm (27.56mils)	PTH	Round	L1 - L4	Pad
□	12	0.73mm (28.74mils)	PTH	Round	L1 - L4	Pad
◇	8	0.75mm (29.53mils)	PTH	Round	L1 - L4	Pad
⊠	20	1.00mm (39.37mils)	PTH	Round	L1 - L4	Pad
✳	1	1.80mm (70.87mils)	PTH	Round	L1 - L4	Pad
⊠	1	2.00mm (78.74mils)	PTH	Round	L1 - L4	Pad
	Total 274					

Drill Drawing L1 - L4 (Scale 1:1)



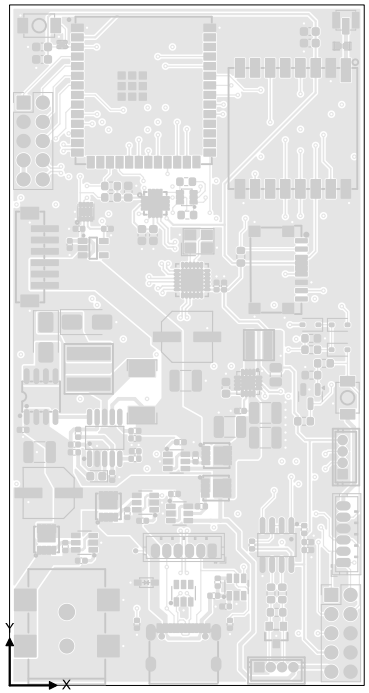
Comments:	Company: Narval Inc.		Variant: PRELIMINARY	Git Hash: 3a63ce1
	Board Name: IndustrialSense		Project Name: IoT NG	
Sheet Title: Drill Drawing (L1 - L4)	File Name: IndustrialSense.kicad_pcb	Designer: William Regalado	Date: 2024-04-13	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: A4	Sheet: 3 of 9

IndustrialSense Fabrication Document

Top Test Points (Scale 1:1)

Ref.	Net	X [mm]	Y [mm]
TP1	VBUS	77.18	-13.40

Ref.	Net	X [mm]	Y [mm]



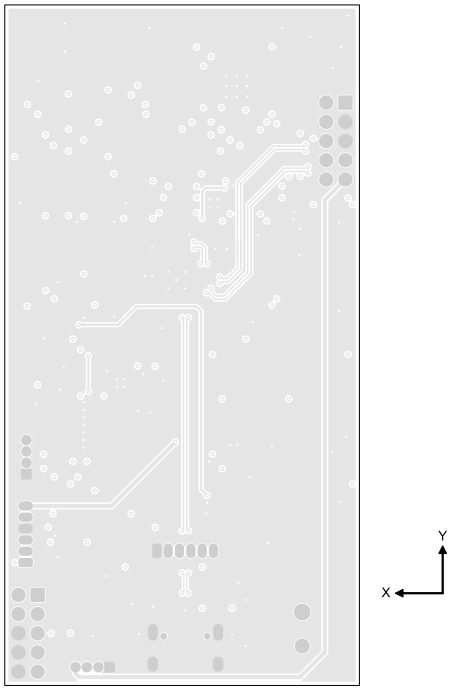
All dimensions are in millimeters unless otherwise specified.

Comments:	Company: Narval Inc.		Variant: PRELIMINARY	Git Hash: 3a63ce1
	Board Name: IndustrialSense		Project Name: IoT NG	
Sheet Title: Top Test Points (Scale 1:1)	File Name: IndustrialSense.kicad_pcb	Designer: William Regalado	Date: 2024-04-13	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: A4	Sheet: 4 of 9

IndustrialSense Fabrication Document

Bottom Test Points (Scale 1:1)

Ref.	Net	X [mm]	Y [mm]

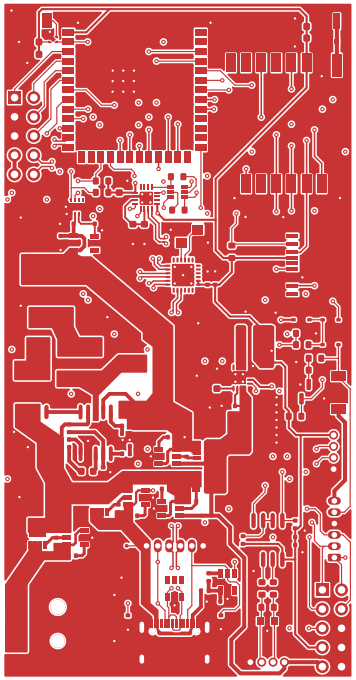


All dimensions are in millimeters unless otherwise specified.

Comments:	Company: Narval Inc.		Variant: PRELIMINARY	Git Hash: 3a63ce1
	Board Name: IndustrialSense		Project Name: IoT NG	
Sheet Title: Bottom Test Points (Scale 1:1)	File Name: IndustrialSense.kicad_pcb	Designer: William Regalado	Date: 2024-04-13	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: A4	Sheet: 5 of 9

IndustrialSense Fabrication Document

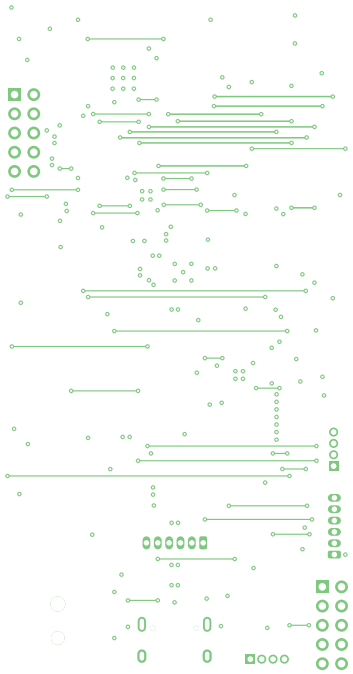
L1 (Scale 1:1)



Comments:	Company: Narval Inc.		Variant: PRELIMINARY	Git Hash: 3a63ce1
	Board Name: IndustrialSense		Project Name: IoT NG	
Sheet Title: L1 (Scale 1:1)	File Name: IndustrialSense.kicad_pcb	Designer: William Regalado	Date: 2024-04-13	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: A4	Sheet: 6 of 9

IndustrialSense Fabrication Document

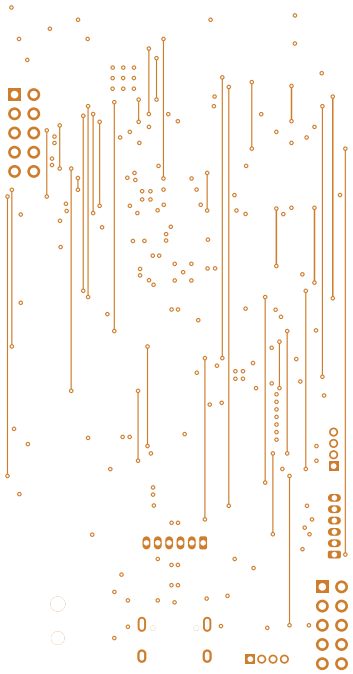
L2 (Scale 1:1)



Comments:	Company: Narval Inc.		Variant: PRELIMINARY	Git Hash: 3a63ce1
	Board Name: IndustrialSense		Project Name: IoT NG	
Sheet Title: L2 (Scale 1:1)	File Name: IndustrialSense.kicad_pcb	Designer: William Regalado	Date: 2024-04-13	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: A4	Sheet: 7 of 9

IndustrialSense Fabrication Document

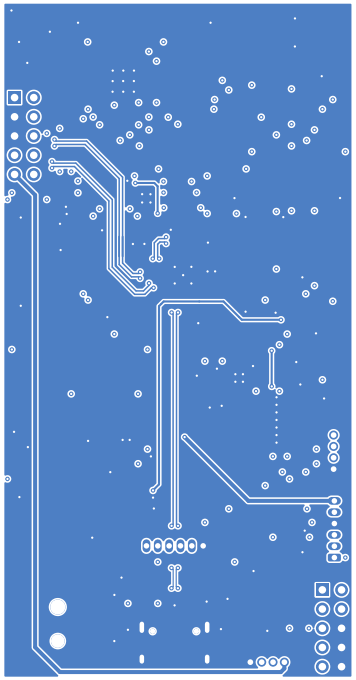
L3 (Scale 1:1)



Comments:	Company: Narval Inc.		Variant: PRELIMINARY	Git Hash: 3a63ce1
	Board Name: IndustrialSense		Project Name: IoT NG	
Sheet Title: L3 (Scale 1:1)	File Name: IndustrialSense.kicad_pcb	Designer: William Regalado	Date: 2024-04-13	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: A4	Sheet: 8 of 9

IndustrialSense Fabrication Document

L4 (Scale 1:1)



Comments:	Company: Narval Inc.		Variant: PRELIMINARY	Git Hash: 3a63ce1
	Board Name: IndustrialSense		Project Name: IoT NG	
Sheet Title: L4 (Scale 1:1)	File Name: IndustrialSense.kicad_pcb	Designer: William Regalado	Date: 2024-04-13	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: A4	Sheet: 9 of 9